

Technical Debt Register and Remediation Plan — KNMP VisitHub

Document Title: Technical Debt Plan **System:** KNMP VisitHub **Author:** Manus AI **Date:** August 2026 **Revision:** 1.0

1. Introduction

This register documents technical debt knowingly accepted during the accelerated development of KNMP VisitHub, rates each item by impact and effort, and proposes a remediation sequence. Items marked **incurred at build** arose from deliberate trade-offs made to ship within the 48-hour development window; items marked **evolved** accumulated as the feature set grew through seven enhancement rounds.

2. Debt Register

ID	Debt Item	Origin	Impact	Effort to Fix	Priority
TD-1	Router file exceeds 400 lines; the recommended split of feature routers into separate files was deferred	Incurred at build	Medium — slower navigation of procedures, merge friction in teams	Low	P2
TD-2	Hardcoded park constants in helpers (park name, contact details, ticket footer) rather than settings-driven values	Incurred at build	Low	Low	P3
TD-3	Capacity and near-capacity threshold were code constants (500 / 75%) until Round 7	Incurred at build	Medium before R7; resolved by site_settings table	—	Resolved
TD-4	Email delivery is fire-and-forget with no retry queue; provider outages silently drop tickets	Incurred at build	Medium — lost tickets during outages	Medium (queue + retry)	P1
TD-5	Multi-day stays are spread evenly across days in the forecast rather than modelled per	Incurred at build	Low — projection smoothing	Medium (slot-level data)	P3

ID	Debt Item	Origin	Impact	Effort to Fix	Priority
	booked slot distribution				
TD-6	Check-in is a booking-level timestamp rather than per-party/per-ticket counting	Incurred at build	Low-medium — gate staff cannot track partial arrivals	Medium (party check-in rows)	P2
TD-7	No API versioning on the tRPC gateway	Evolved	Low — in-app clients only today	Medium (versioned routers)	P3
TD-8	Analytics queries read all matching rows to aggregate instead of pre-aggregated rollups	Evolved	Low-medium at current scale; will degrade with years of history	Medium (materialised tables)	P2
TD-9	Currency arithmetic is integer pesewas but display helpers are scattered across components	Evolved	Low — formatting drift risk	Low (shared formatter)	P2
TD-10	TiDB prepared-statement incompatibility with SELECT aliases in ORDER BY required driver-level workaround	Resolved in R4	Resolved — positional ordering + regression tests	—	Resolved
TD-11	No seeded demonstration data tooling; database seed is	Incurred at build	Low	Low (seed script)	P3

ID	Debt Item	Origin	Impact	Effort to Fix	Priority
	migration-time only				
TD-12	Rate limiting absent on public endpoints (search, capacityCheck, share lookups)	Incurred at build	Medium — abuse surface on public routes	Medium (edge or app-level limiting)	P1
TD-13	QR check-in has no offline tolerance; camera scan requires network	Incurred at build	Medium for gate operations	High (offline queue/sync)	P2

3. Resolution Narrative

Two significant items have already been retired. **TD-3** was resolved in Round 7 when park operations gained a Settings page backed by the `site_settings` table, with sensible schema constants (500 visitors, 75%) as fallbacks. **TD-10** was resolved in Round 4: the prepared-statement incompatibility was fixed with positional ordering, and three live-database regression tests guard against recurrence.

4. Remediation Plan

The plan is sequenced by priority, assuming the current architecture is retained.

Wave 1 — Reliability (P1, recommended next). TD-4 should be addressed first because lost ticket emails directly affect visitor experience. A small durable outbox table with a scheduled retry job (using the platform’s heartbeat scheduling) would guarantee delivery. TD-12 follows: per-IP rate limiting on public endpoints, ideally at the edge, protects the park from scraping and denial of service.

Wave 2 — Operational fidelity (P2). TD-6 upgrades gate check-in to per-party attendance counting, which pairs naturally with a paper-free daily manifest feature. TD-8 adds monthly revenue and visitor rollups so analytics remain fast as history grows. TD-9 consolidates currency display into a shared formatter. TD-1 is a housekeeping split of `routers.ts` into per-feature router files, safe because the caller contract is unchanged. TD-13 is a longer-horizon item requiring an offline-first gate client.

Wave 3 — Hygiene (P3). TD-5, TD-7, TD-2, and TD-11 are low-urgency improvements that can be folded into future feature work: slot-level attendance modelling, versioned routers, settings-driven static content, and a repeatable seed fixture.

5. Preventive Practices

To avoid compounding the register, ongoing work should keep new procedures in feature-scoped router files, extend `site_settings` rather than adding constants for operational values, cover new SQL with live-database tests, and treat email-sending code paths as retryable by default. Each checkpoint should continue to require a green Vitest run before delivery.